Fastmixture

2024-11-14

```
res_dir <- "../results/s05_fastmixture/"
```

This script uses fastMIXTURE (Santander et al., 2024) through the package tidypopgen (Carter and Manica, 2025) and clumppling (ref) through the package tidygenclust (ref) to estimate ancestry proportions of individuals in the global dataset.

Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt",
    reattach_to = "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean"
  )
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 1178
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 498926
```

fastMIXTURE was run on the full, cleaned dataset of 1178 individuals and 498926 SNPs.

```
if (file.exists("../data/intermediate/fastmixture/fastmixture_m.RData")) {
  qmats <-
    readRDS("~/Dropbox/gitdata/malagasy_diversity/data/intermediate/fastmixture/fastmixture_m.RData")
} else {
  tictoc::tic()
  k <- c(2:10)
  qmats <- gt_fastmixture(mad_bor_g1000_sum,
    k,
    n_runs = 3,
    threads = 1,
    batches = 16,
    seed = c(1, 2, 3),
    no_freqs = TRUE
  )
  saveRDS(qmats, "../data/intermediate/fastmixture/fastmixture_m.RData")
  time <- tictoc::toc()
}
```

fastMIXTURE was run unsupervised for k values `min(range(unique(qmats$k)))` to 10, with 3 repeats.

```
# check order
testthat::expect_equal(qmats$id, mad_bor_g1000_sum$id)
# add grouping variable
qmats$group <- mad_bor_g1000_sum$population
```

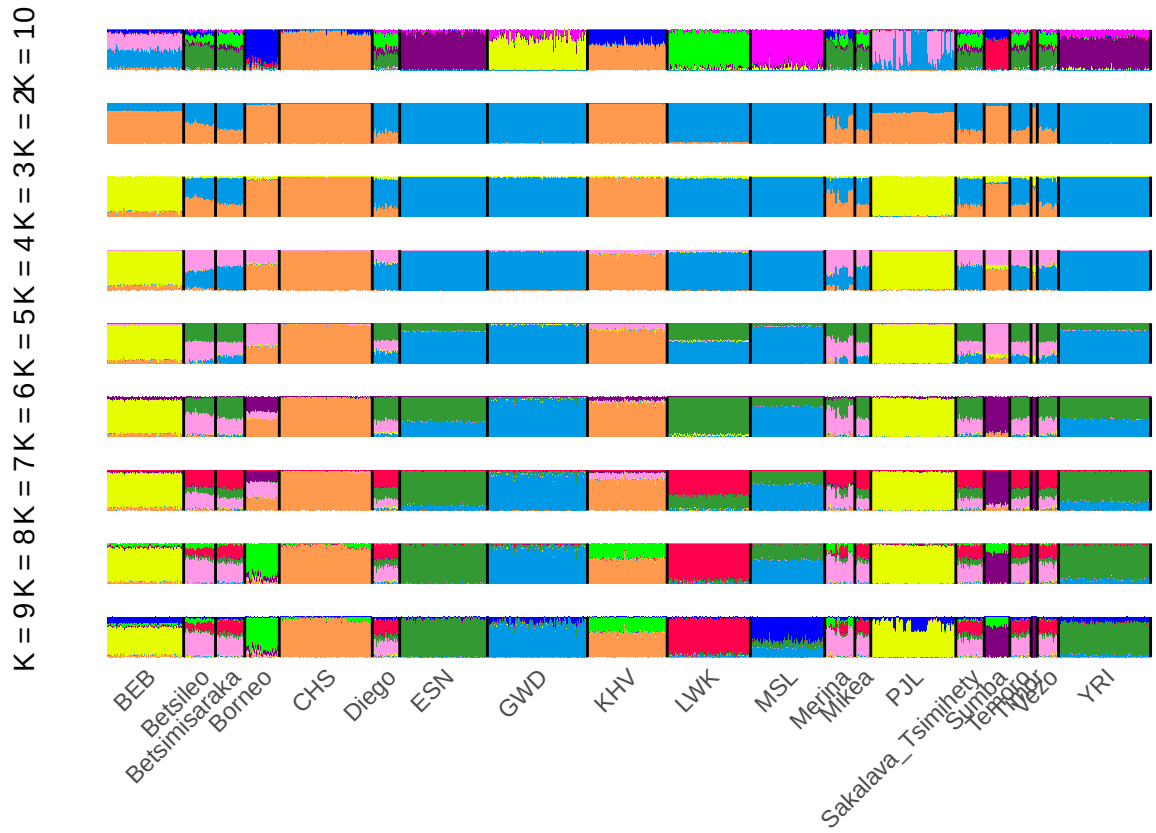
Reorder the gt_admix object

```
qmats <- gt_admix_reorder_q(qmats)
```

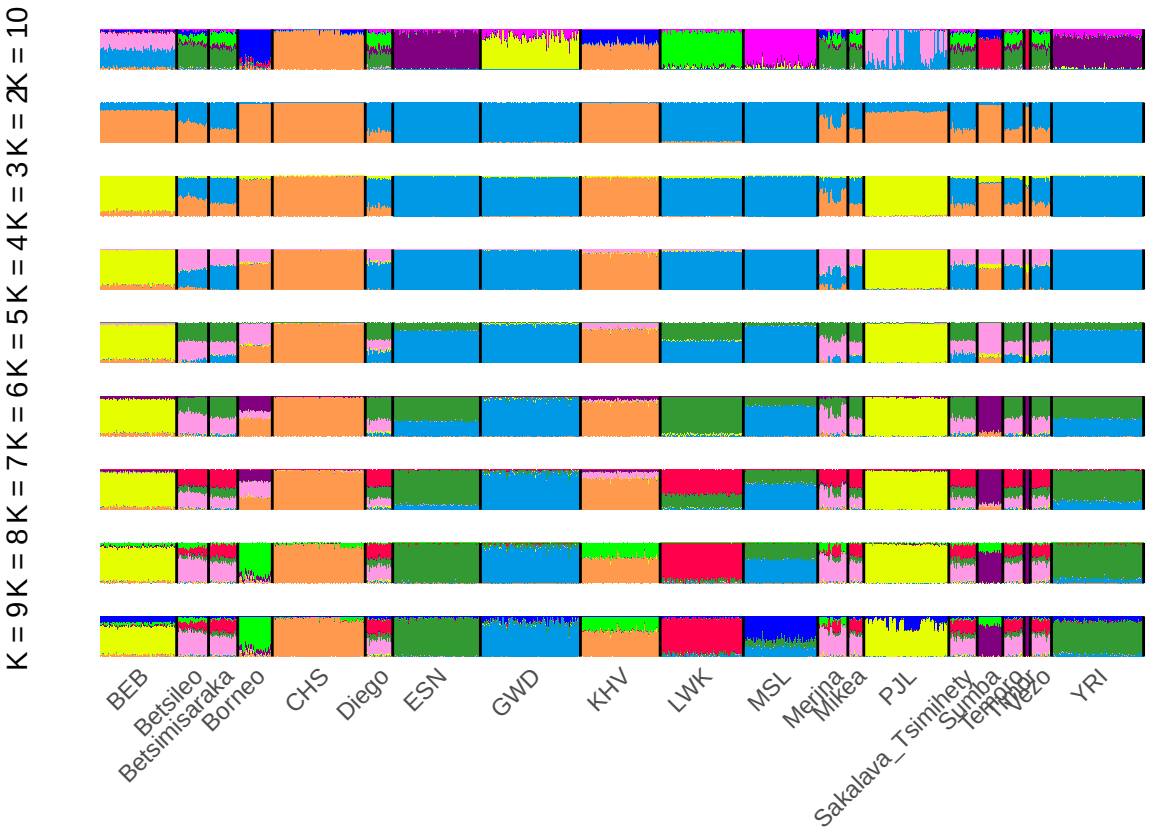
Clumppling

```
clump <- gt_clumppling(input_path = qmats)
```

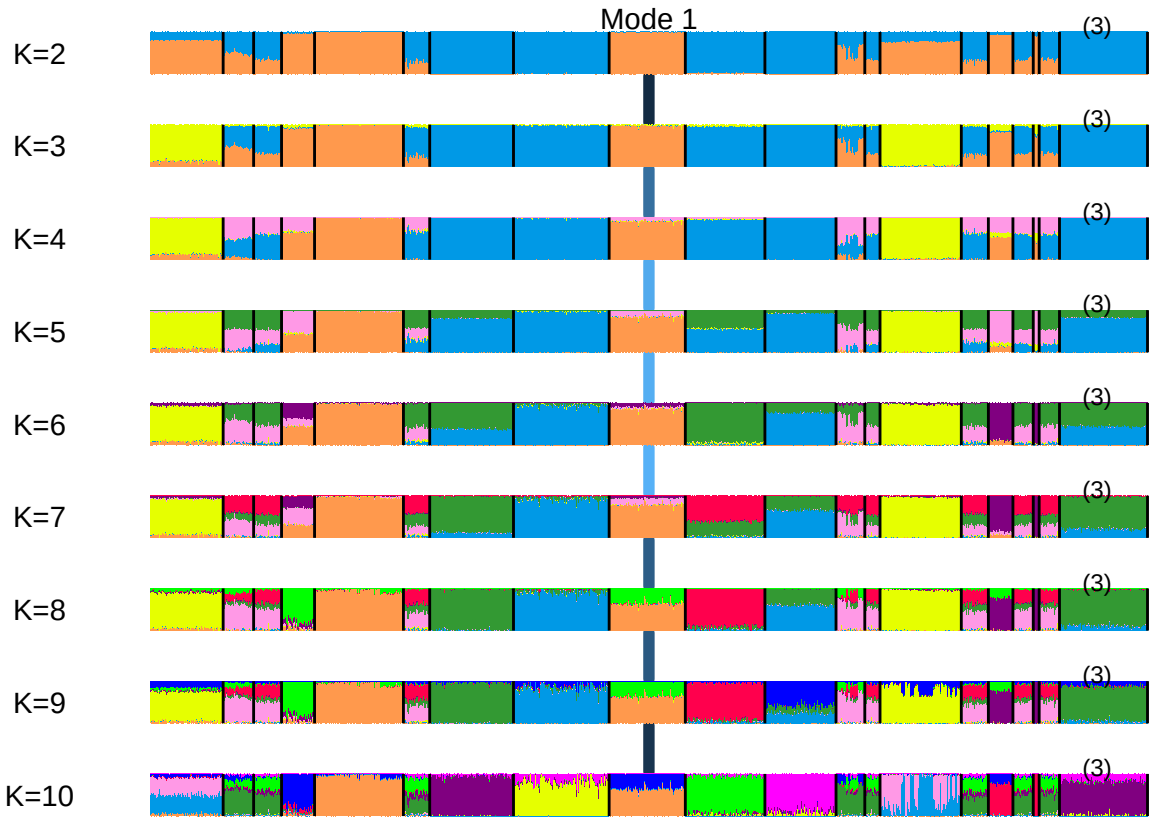
```
clump %>% autoplot(type = "all_modes", group = qmats$group)
```




```
clump %>% autoplot(type = "major_modes", group = qmats$group)
```

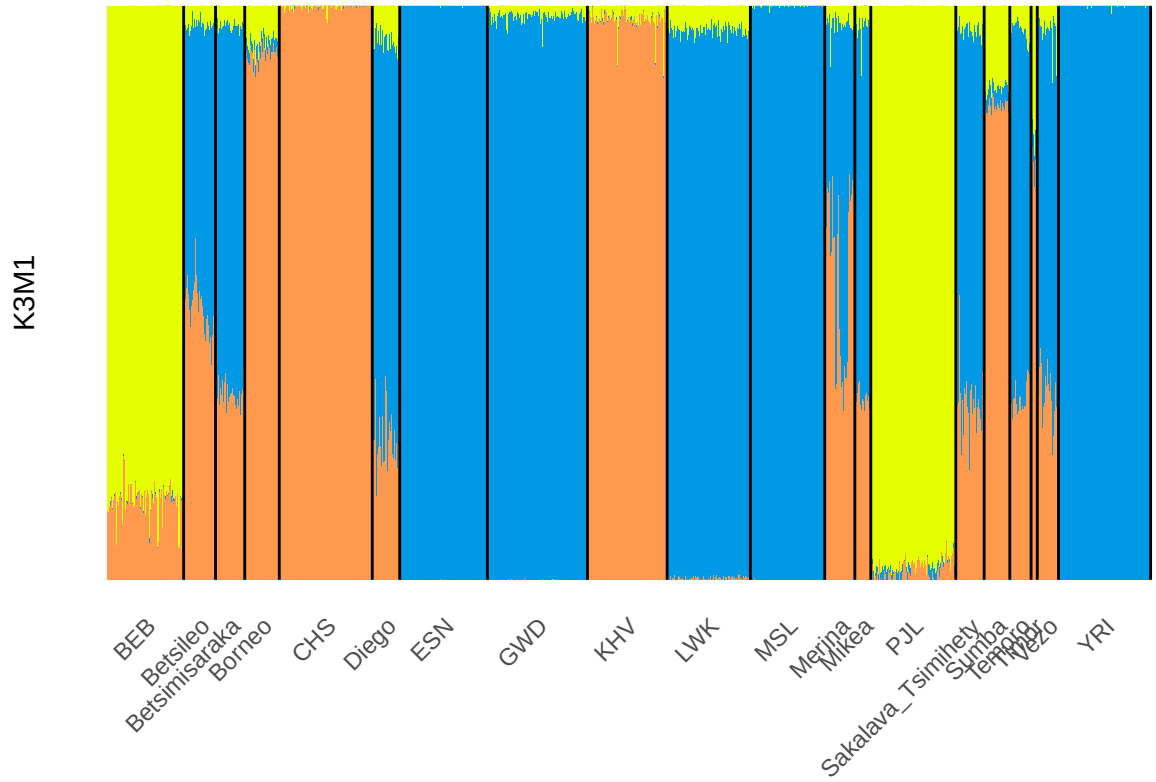



```
clump_out <- clump %>% autoplot(type = "modes", group = qmats$group)
clump_out
```



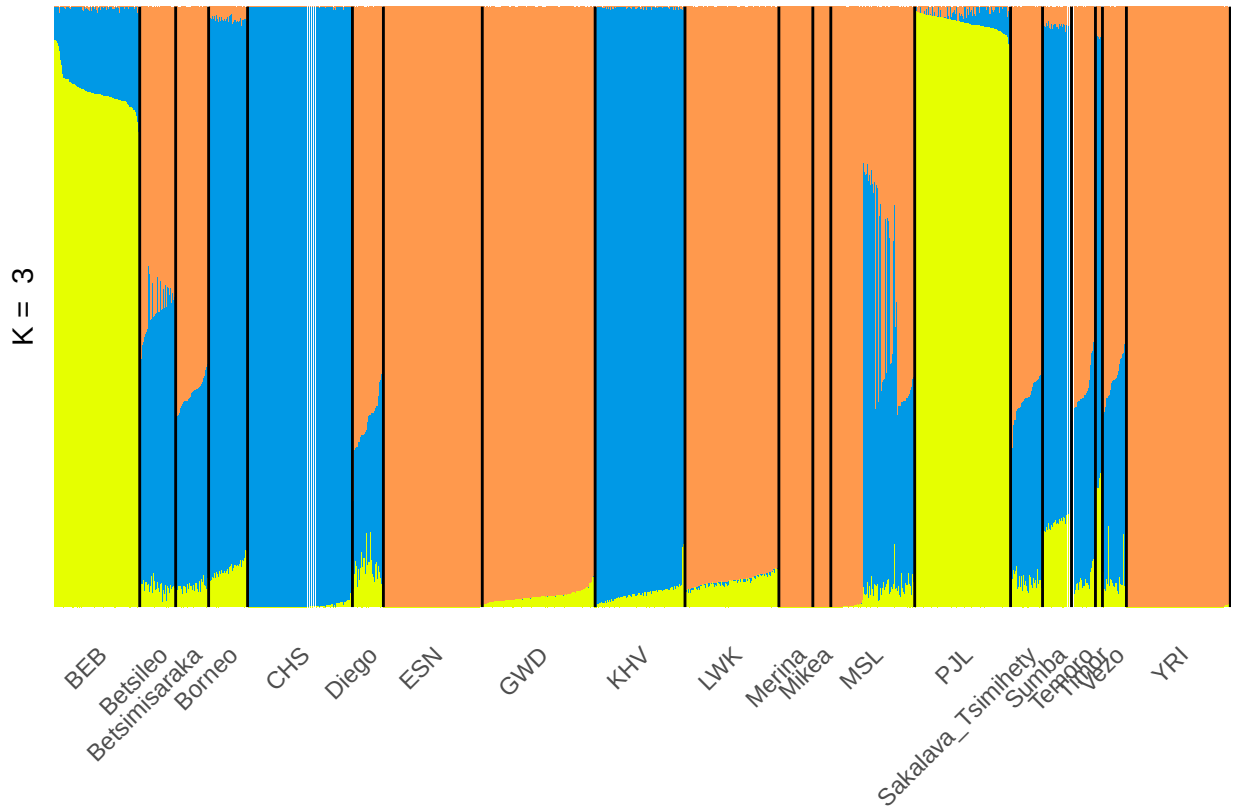

```
ggsave(paste0(res_dir, "clumppling_modes.jpg"),
       width = 7, height = 7, dpi = 300)
```

```
clump %>% autoplot(type = "modes_within_k", k = 3, group = qmats$group)
```



Initial plot

```
autoplot(get_q_matrix(qmats, k = 3, run = 1),
  data = mad_bor_g1000_sum,
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
)
```



Add full population names

```
pop_names <-
  read.csv("../data/raw/data_input/igsr-1000-genomes-phase-3-release.tsv",
    sep = "\t"
  )

# find matching id in pop_names Sample.name and mad_bor_g1000_sum id, and for the matches, replace the
for (i in seq_len(nrow(pop_names))) {
  if (pop_names$Sample.name[i] %in% mad_bor_g1000_sum$id) {
    mad_bor_g1000_sum[mad_bor_g1000_sum$id == pop_names$Sample.name[i], "population"] <- pop_names$Popul
  }
}

# replace "Luhya,Luhya" with "Luhya"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Luhya,Luhya"] <- "Luhya"
# replace "Esan,Esan" with "Esan"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Esan,Esan"] <- "Esan"
# replace "Gambian Mandinka,Gambian" with "Gambian Mandinka"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Gambian Mandinka,Gambian"] <- "Gambian Ma
# replace "Mende,Mende" with "Mende"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Mende,Mende"] <- "Mende"
# replace "Kinh,Kinh Vietnamese" with "Kinh Vietnamese"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Kinh,Kinh Vietnamese"] <- "Kinh Vietnames
# replcae "Punjabi,Punjabi" with "Punjabi"
```

```
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Punjabi,Punjabi"] <- "Punjabi"
# replcae "Benga,Bengali" with "Bengali"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Bengali,Bengali"] <- "Bengali"

# replace any _ with a space
mad_bor_g1000_sum$population <- gsub("_", " ", mad_bor_g1000_sum$population)
```

```
# reorder
mad_bor_g1000_sum$Region <- "Other"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <- "Island
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Bengali", "Punjabi"), "Region"] <- "South Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Southern Han Chinese", "Kinh Vietnamese"), "Regi
# Not really happy about grouping all African samples here though...
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Esan", "Gambian Mandinka", "Luhya", "Mende", "Yo
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Betsimisaraka", "Merina", "Diego", "Sakalava Tsi

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% arrange(Region, population)
```

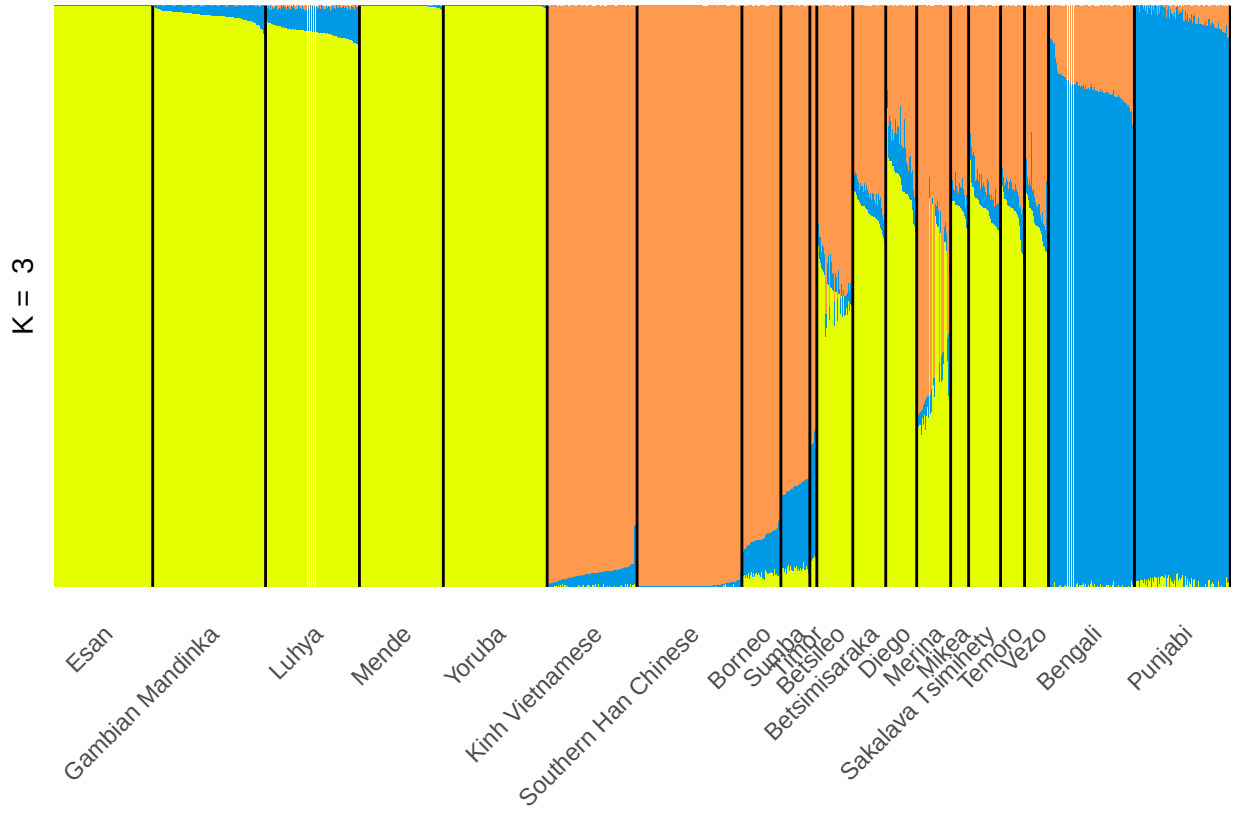
Arrange and save plot

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% arrange(Region, population)
pop_levels <- unique(mad_bor_g1000_sum$population)

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% filter(.data$id %in% qmats$id)
mad_bor_g1000_sum <- mad_bor_g1000_sum[match(qmats$id, mad_bor_g1000_sum$id), ]
pops <- mad_bor_g1000_sum$population
pops <- factor(pops, levels = pop_levels)

qmats <- gt_admix_reorder_q(qmats, group = pops)

autoplot(get_q_matrix(qmats, k = 3, run = 1),
  annotate_group = TRUE, arrange_by_group = TRUE,
  arrange_by_indiv = TRUE, reorder_within_groups = TRUE
)
```




```
ggsave(paste0(res_dir, "fastmixture_autoplot.jpg"),
       width = 7, height = 7, dpi = 300)
```

Check which ancestry components the .Q files represent

```
augmented <- augment(get_q_matrix(qmats, k = 3, run = 1),
  data = mad_bor_g1000_sum
)

# filter augmeneted to $population GWD and print head
augmented_gwd <- augmented %>% filter(population == "Gambian Mandinka")
head(augmented_gwd)
```

```
## # A tibble: 6 x 9
## # Groups:   population [1]
##   id      population      genotypes Region .rownames      .Q1    .Q2      .Q3 group
##   <chr>   <chr>           <vctr_SN> <chr>      <int>    <dbl> <dbl>    <dbl> <int>
## 1 HG02461 Gambian Mandin~       477 Africa       430 1.04e-3 0.982 0.0167       2
## 2 HG02462 Gambian Mandin~       478 Africa       431 1.00e-5 0.960 0.0401       2
## 3 HG02464 Gambian Mandin~       479 Africa       432 1.00e-5 0.998 0.00210      2
## 4 HG02465 Gambian Mandin~       480 Africa       433 1.00e-5 0.991 0.00912      2
## 5 HG02561 Gambian Mandin~       489 Africa       434 1.23e-4 0.967 0.0330       2
## 6 HG02562 Gambian Mandin~       490 Africa       435 1.00e-5 0.982 0.0180       2
```

```r
# filter augmented to $population Borneo and print head
augmented_borneo <- augmented %>% filter(population == "Borneo")
head(augmented_borneo)
```

```
## # A tibble: 6 x 9
## # Groups:   population [1]
##   id     population  genotypes Region        .rownames   .Q1    .Q2    .Q3 group
##   <chr>  <chr>      <vctr_SNP> <chr>             <int> <dbl>  <dbl>  <dbl> <int>
## 1 BJM003 Borneo            234 Island south~       156 0.918 0.0141 0.0680     8
## 2 BJM024 Borneo            235 Island south~       157 0.908 0.0249 0.0675     8
## 3 BJM030 Borneo            236 Island south~       158 0.922 0.0236 0.0545     8
## 4 BJM034 Borneo            237 Island south~       159 0.908 0.0224 0.0697     8
## 5 BJM038 Borneo            238 Island south~       160 0.902 0.0213 0.0770     8
## 6 BJM061 Borneo            239 Island south~       161 0.898 0.0201 0.0818     8
```

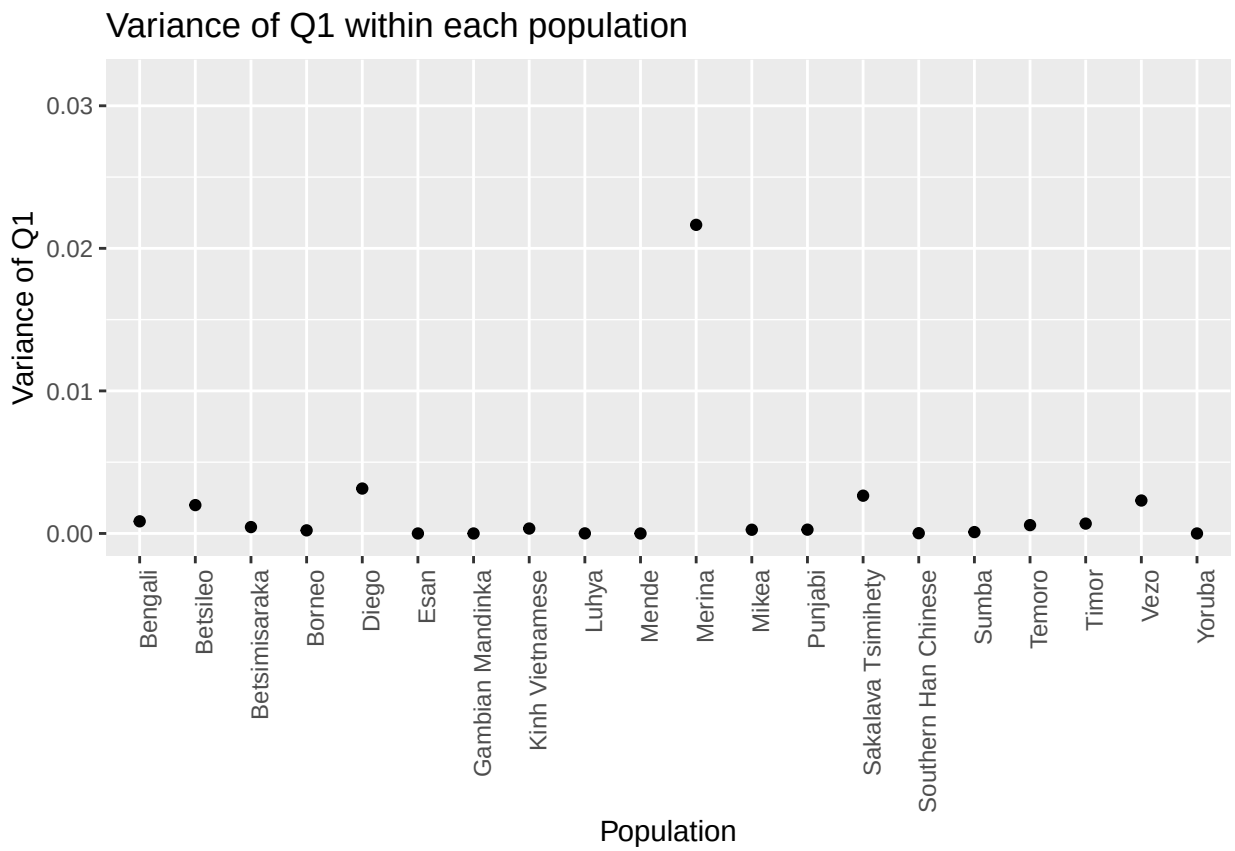
```r
# .Q3 representing Asian ancestry and .Q2 representing African ancestry
```

Look at variance within each population

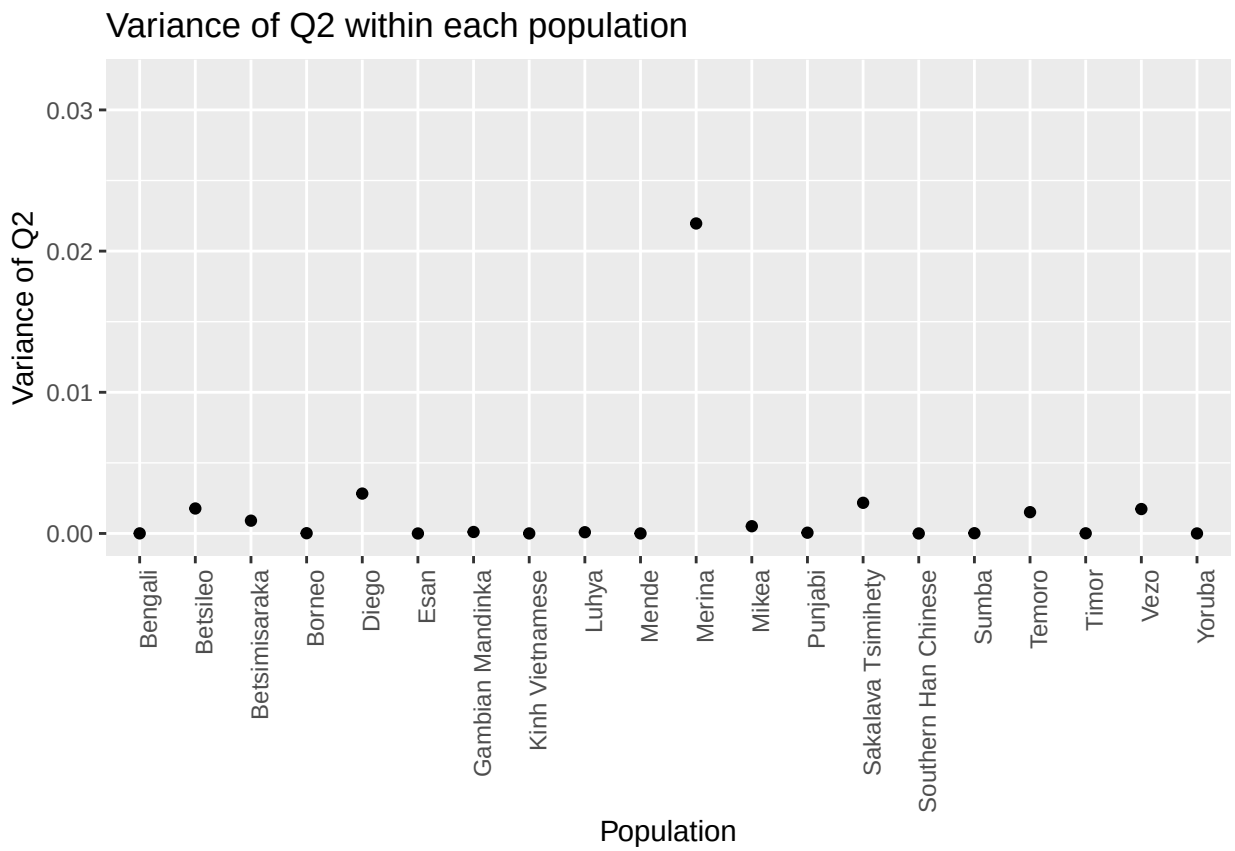
```r
vars <- augmented %>%
  group_by(population) %>%
  summarise(Q1 = var(.Q1), Q2 = var(.Q2), Q3 = var(.Q3))
head(vars)
```

```
## # A tibble: 6 x 4
##   population              Q1          Q2           Q3
##   <chr>                <dbl>       <dbl>        <dbl>
## 1 Bengali       0.000852     0.00000746  0.000869
## 2 Betsileo      0.00199      0.00177     0.000103
## 3 Betsimisaraka 0.000449     0.000907    0.000509
## 4 Borneo        0.000219     0.0000180   0.000193
## 5 Diego         0.00315      0.00283     0.000486
## 6 Esan          0.0000000298 0.000000205 0.0000000988
```

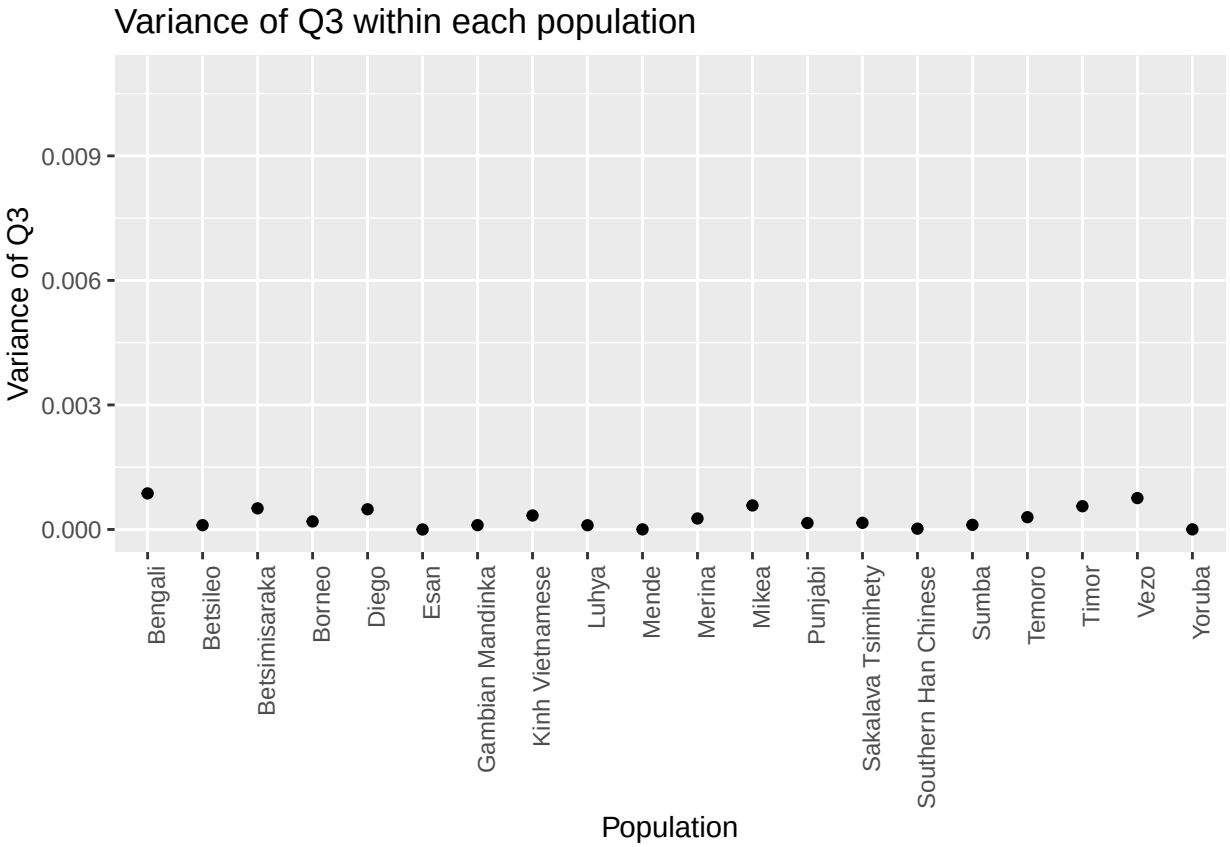
```r
ggplot(vars, aes(x = population, y = Q1)) +
  geom_point() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Variance of Q1 within each population",
    x = "Population", y = "Variance of Q1"
  ) +
  scale_y_continuous(limits = c(0, max(vars$Q1) + 0.01))
```




```r
ggplot(vars, aes(x = population, y = Q2)) +
  geom_point() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Variance of Q2 within each population",
    x = "Population", y = "Variance of Q2"
  ) +
  scale_y_continuous(limits = c(0, max(vars$Q2) + 0.01))
```




```
ggplot(vars, aes(x = population, y = Q3)) +
  geom_point() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Variance of Q3 within each population",
    x = "Population", y = "Variance of Q3"
  ) +
  scale_y_continuous(limits = c(0, max(vars$Q3) + 0.01))
```


Variance of Q3 within each population
Variance of Q3
0.009
0.006
0.003
0.000
Bengali
Betsileo
Betsimisaraka
Borneo
Diego
Esan
Gambian Mandinka
Kinh Vietnamese
Luhya
Mende
Merina
Mikea
Punjabi
Sakalava Tsimihety
Southern Han Chinese
Sumba
Temoro
Timor
Vezo
Yoruba
Population